

A	M	T	W	T	F	S	S
P	1	2	3	4	5	6	7
R	8	9	10	11	12	13	14
	15	16	17	18	19	20	21
	22	23	24	25	26	27	28
'19	29	30					

MARCH Wk. - 10
Saturday 068-297

09

Nested for loop:-

1. If total row and columns are not same and first value is changing and last value is fixed and first value is lowest and last value is highest then always $j=i$, $j \leq \text{highest value}$. j is incrementing. (i is always changing).

Ex →

	1	2	3	4
1	2	3	4	
	3	4		
2	4			

2. If total row and columns are not same and first value is changing and last value is fixed and first value is highest and last value is lowest then $j=i$, $j \geq \text{lowest value}$. j is decrementing. (i is changing)

Ex →

	4	3	2	1
	3	2	1	
Evening	2	1		
	1			

Sunday 10

3. If total row and columns are not same. First value is fixed and last value is changing and first value is lowest and last value is highest then $j = \text{lowest value}$ (fixed value), $j \leq i$. j is incrementing.

Wk. - 11 MARCH
070-295 Monday

M	T	W	T	F	S	S	
				1	2	3	M
4	5	6	7	8	9	10	A
11	12	13	14	15	16	17	R
18	19	20	21	22	23	24	
25	26	27	28	29	30	31	'19

Ex → 1 2 3 4

9 1 2 3

1 2

10

1

- 4) ¹¹ If total row and columns are not same and first value is fixed and last value is changing ¹² and first value is highest and last value is lowest than $j = \text{first value (highest value)}$, $j \geq i$ ¹ j is decrementing. (i is changing).

2 Ex → 4 3 2 1

4 3 2

3

4 3

4

4

- 5) If total number of row and column are same ⁵ then there is no relationship between i and j (row and column) that means i and j both loop ⁶ will execute same number of times j is not dependent on i .

Evening

Ex → 1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

	M	T	W	T	F	S	S
A	1	2	3	4	5	6	7
P	8	9	10	11	12	13	14
R	15	16	17	18	19	20	21
	22	23	24	25	26	27	28
'19	29	30					

MARCH Wk - 11
Tuesday 071-294

12

6) If only a single value present within the first row then that must be the i's value and j is depends on i.

Ex →

1 2

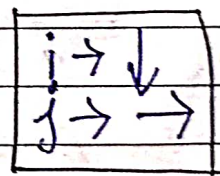
1 2 3

1 2 3 4

Note:—

If within the same row value not j's value, otherwise print j's value.

Syntax:



Outer for loop
for ()
{
 for ()
 {
 // inner for loop
 }
}

Note:—

Inner for loop will execute more times than outer for loop.

5) 1
12
123
1234

6) 1234
1234
1234
1234

7) 13579
1357
135
13
1

8) 1
31
531
7531
97531

9) 9
9 7
9 7 5
9 7 5 3
9 7 5 3 1

10) 9
7 9
5 7 9
3 5 7 9
1 3 5 7 9

11) 1
2 1
3 2 1
4 3 2 1

12) 4
3 4
2 3 4
1 2 3 4

13) 1
2 2
3 3 3
4 4 4 4

14) 4
3 3
2 2 2
1 1 1 1

15) 4 4 4 4
3 3 3
2 2
1

16) 5
4 4
3 3 3
2 2 2 2
1 1 1 1 1

17) * * * *
 * * *
 * *
 *

18) *
 * *
 * * *
 * * * *

19) = * * *
 * = * *
 * * = *
 * * * =

20) = - - *
 - - * *
 - * * *
 * * * *

21) $\begin{array}{cccc} * & * & * & * \\ - & * & * & * \\ - & - & * & * \\ - & \cancel{*} & \cancel{*} & * \\ \hline \end{array}$

22) $\begin{array}{ccccccc} - & - & - & * & & & \\ - & - & * & & * & & \\ - & * & & * & & * & \\ * & & * & & * & & * \end{array}$

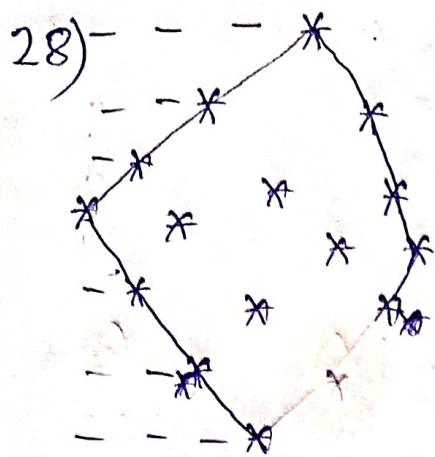
23) $\begin{array}{cccc} * & * & * & * \\ - & * & * & * \\ - & - & * & * \\ - & - & - & * \end{array}$

24) 1
0 1
1 0 1
0 1 0 1
1 0 1 0 1

25) 0
1 1
0 0 0
1 1 1 1

26) 1
1 0
1 0 1
1 0 1 0

27) 1
0 0
1 1 1
0 0 0 0
1 1 1 1 1



29)

1						
1	2	3				
1	2	3	4	5		
1	2	3	4	5	6	7

30)

1						
1	2	3				
1	2	3	4	5		
1	2	3	4	5	6	7

31)

1						
1	2	1				
1	2	3	2	1		
1	2	3	4	3	2	1

32)

1						
1	2	1				
1	2	3	2	1		
1	2	3	4	3	2	1

33)

*	*	*				
*	*	*	*	*	*	
*	*	*	*	*	*	*

34)

*	*	*	*	*	*	*
*	*	*	*	*	*	*
*	*	*	*	*	*	*
*	*	*	*	*	*	*

35)

*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*

36)

*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*
*	*	*	*	*	*	*	*	*	*

37)

1						
1	1					
1	1	2				
1	1	2	3			

38)

1						
2	6					
3	7	10				
4	8	11	13			
5	9	12	14	15		

39)

1	2	3	4	5	3	2	1
1	2	3	*	*	3	2	1
1	2	*	*	*	*	2	1
1	*	*	*	*	*	*	1

40)

1						
2	3	4				
5	6	7	8	9		

41)

-	-	-	-	1		
-	-	-	2	1		
-	-	3	2	1		
-	4	3	2	1		
5	4	3	2	1		

42)

1						
2	3	2				
4	5	6	5	4		
7	8	9	10	9	8	7

43) 1 - - - - - P
 1 2 - - - - - 21
 1 2 3 - - - - 3 2 1
 1 2 3 4 - - 4 3 2 1
 1 2 3 4 5 4 3 2 1

44) 1
 2 1 2
 3 2 1 2 3
 4 3 2 1 2 3 4

45) 1
 2 3
 3 4 5
 4 5 6 7
 5 6 7 8 9

46) A A A A
 A A A
 A A
 A

47) A
 A A
 A A A
 A A A A

48) A B C D
 B C D
 C D
 D

47) A B C D
 A B C
 A B
 A

48) A
 B C
 D E F
 G H I J

49) A B C D
 E F G
 H I
 J

50) A B C D
 E F G H
 I J K L
 M N O P

51) A
 A B
 A B C
 A B C D

52) A
 B B
 C C C
 D D D D

53) D
D C
D C B
D C B A

54) D C B A
B D C B
D C - - - - -
D - - - - -

54) D C B A

55) D C B A
C B A
B A
A

56) D
C C
B B B
A A A A

57) D D D D
C C C
B B
A

~~54~~ 58) A A A A
B B B
C C
D